

THE BUS HARNESS

The PICO9918 PRO sits in the TMS9918A's 40-pin DIP position on a breadboard or a DIP-40 socket. An Arduino Nano drives its four ports; the Mac drives the Nano over USB serial. Every connection below is a jumper wire. Nothing is fabricated.

PREPARING THE PRO

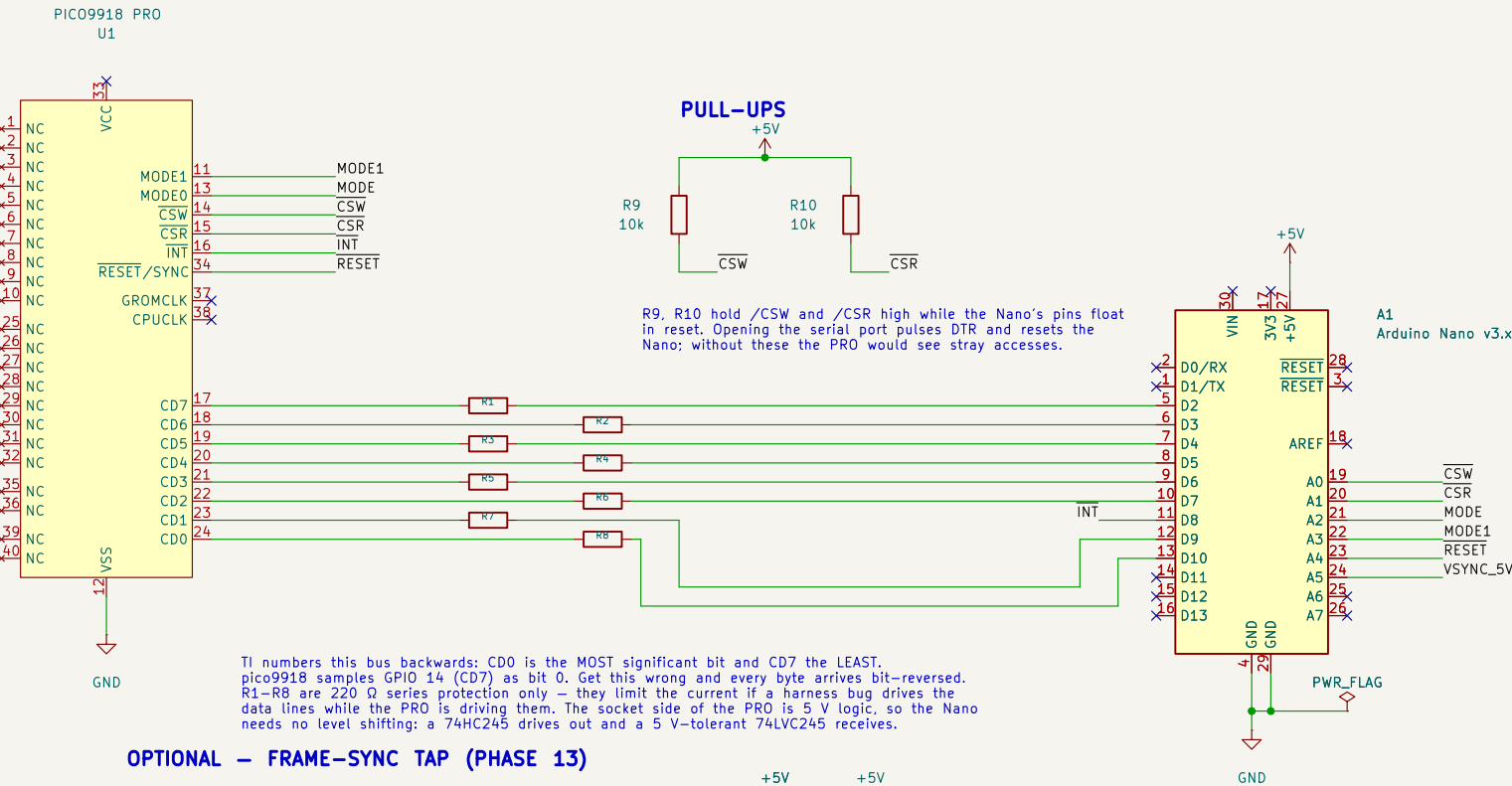
MDE1 (pin 11) has no pin fitted: MODE1 is not a TMS9918 signal, so the PRO ships with that pad bare. Solder one 0.1" male header pin into it. Without it, port B (\$9C02/\$9C03) is unreachable — on the bench and in the AC6502 alike. No other modification is needed.

Pin 33 (+5V) is LEFT UNCONNECTED. The PRO runs from its own USB-C, which also carries the debug link to the Mac. The Nano runs from its own USB. Only ground is common: tie the Nano's GND to the PRO's pin 12 through the breadboard's ground rail.

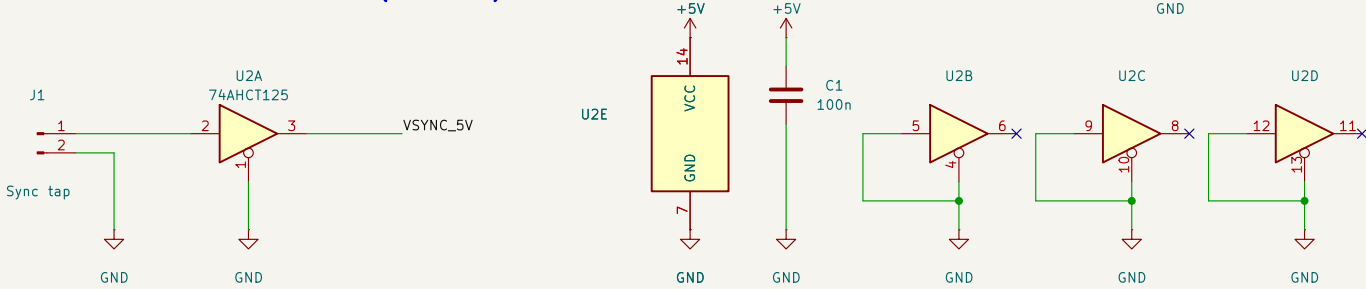
Back up the stock flash before anything else: picotool save, in BOOTSEL, kept outside the repo.

PORT DECODE

MODE1:MODE = CPU A1:A0. They select the port:
0 0 \$9C00 data 0 1 \$9C01 address / register
1 0 \$9C02 port B data 1 1 \$9C03 port B command



OPTIONAL — FRAME-SYNC TAP (PHASE 13)



Not fitted by default. With the VGA dongle the sync source is the DE-15's VSYNC (pin 14) and GND (pin 10). With the HDMI dongle there is no analogue sync to tap; bring frame sync out on a spare PRO GPIO in this repo's firmware instead, or leave the tap off and time frames from /INT alone. U2 translates a 3.3 V source to 5 V. HCT thresholds (VIH 2.0 V) make 3.3 V an unambiguous high; the Nano's own VIH of 3.0 V does not. A5 = PC5 = PCINT13, timestamped inside its pin-change interrupt.

THE NANO

D0 and D1 carry the USB serial link to the Mac at 1,000,000 baud (500,000 is the exact fallback). The 328P's 16 MHz clock hits both rates exactly; the USB bridge and its macOS driver must too.

D0/D1 are taken by that link, and PB6/PB7 and PC6 hold the crystal and reset, so the 328P has no free 8-bit port. Data is split: six bits on PORTD, two on PORTB. Control stays on PORTC alone, so each strobe edge is still a single port write. On a write, both data ports are set before /CSW falls. On a read, both are sampled while /CSR is low, one instruction (62.5 ns) apart, and the PRO holds the bus throughout.

D8 = PB0 = ICP1, the 328P's ONLY input-capture pin. /INT must land here: Timer 1 timestamps its edges to 62.5 ns.

D11-D13 stay free. D13's on-board LED is the harness's status light.

VIDEO

The PRO's video leaves by the FFC cable to its dongle, and never touches this harness: PRO → FFC → HDMI dongle → HDMI capture card → Mac 'vdpctl grab' saves a PNG through ffmpeg -f avfoundation. Capture is never the pass/fail oracle — the snapshot over the debug link is. Capture proves what a snapshot cannot see: sync lock, DAC bit order, line doubling, a picture that stays stable under load.

SIGNAL TIMING

The harness runs three profiles, set from the host:
6502-1mhz -500 ns strobes, accesses $\geq 4 \mu\text{s}$ apart
6502-2mhz -250 ns strobes, accesses $\geq 2 \mu\text{s}$ apart
fastest as fast as the 328P can drive the pins
Strobe width, data setup and hold can also be set directly, for margin sweeps. The harness never asserts /CSR and /CSW together, and tri-states both data ports before /CSR falls.

See docs/BENCH.md for the wiring table, the harness protocol and the bring-up order.

Breadboard wiring reference — not a fabricated board
PICO9918 PRO v2.0 driven by an Arduino Nano, for 6502-PICOVDP Phase 9

A.C. Wright Design

Sheet: /
File: picovdp-bench.kicad_sch

Title: PICOVDP Bench Harness

Size: A3	Date: 2026-09-23	Rev: 1.0
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